

Dr. Vahram Martirosyan

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Abu Dhabi, UAE

Chief Scientist & Machine Learning Lead · Explainable AI (PRMs) · Edge/On-Device AI · HPC & Compiler Engineering · PhD Computational Mathematics

Professional Summary

Ph.D. in Computational Mathematics with **20+ years of experience** spanning explainable AI research, high-performance computing, edge/on-device deployment, and software engineering. Inventor and lead researcher behind **Polynomial Regression Models (PRMs)** — a family of intrinsically interpretable models that deliver results competitive with deep learning at a fraction of the computational cost, across computer vision, healthcare (ECG and clinical diagnosis), materials science, agriculture, and ocean/climate monitoring.

Currently **Chief Scientist** at [ExplainableAI.app](#), leading the full scientific and technical strategy of a no-code Explainable AI platform: browser-side WebAssembly training (data never leaves the device), a hand-written C++ inference engine with sub-millisecond latency, and a Gemini-powered AI App Builder that turns any trained model into a shareable live web app. Research translated into deployable systems: models that run in **1.09 ms on an ESP32**, a **657 KB satellite oil-spill detector** processing 11,500 tiles/s, and a **99.7% model compression** pipeline for 123M-parameter ECG classifiers. Author of **25+ publications** and co-inventor on a pending patent. Former head of a Russian Academy of Sciences laboratory and serial technical leader across research and industry.

Technical Skills

- **Machine Learning & Data Science:** Supervised learning, regression modelling, interpretable model design, feature engineering, model compression/sparsification, cross-validation, Python, PyTorch, scikit-learn, Statistics & Probability.
- **Explainable AI (PRMs):** Polynomial Regression Models, stagewise and L2/elastic-net regularization, feature importance, interpretable classification across vision, tabular, time-series, and remote-sensing data; no-code XAI platform design and deployment.
- **Edge / On-Device AI:** int8/int16 post-training quantization, fixed-point kernels, tiny models (sub-MB) for microcontrollers and edge hardware (ESP32, RP2350, Arduino), latency/energy budgeting, on-board anomaly monitoring.
- **HPC & Parallelism:** LLVM/Clang (compiler pass development), binary lifting, OpenCL, MPI, POSIX Threads, auto-vectorisation (SIMD), heterogeneous CPU-GPU execution, Xilinx ACAP.
- **Systems & Inference Engineering:** C/C++ (expert), WebAssembly, sub-millisecond inference engines, Linux kernel modules, device drivers (Ext4, XFS, JFS, Btrfs).
- **Web & Databases:** PHP/MySQL, ASP.NET/C#/MS SQL Server, NoSQL, SQL.
- **Tooling:** Git, Linux/Unix, L^AT_EX.
- **Languages:** Armenian (native), Russian (fluent), English (fluent).

Experience

Chief Scientist

Jul 2025 – Present

ExplainableAI.app

- Leading the scientific and technical vision of [ExplainableAI.app](#) — a no-code Explainable AI platform built entirely on intrinsically interpretable Polynomial Regression Models, with zero black-box components.
- Designed the full ML pipeline: browser-side WebAssembly training (data never leaves the device), a hand-written C++ inference engine with sub-millisecond latency, and seamless deployment of models as shareable live web apps via a Gemini-powered AI App Builder.
- Platform supports any domain — users train interpretable models on CSV or image datasets and deploy them instantly, without writing code or managing infrastructure.
- Authored 15+ research papers in 2026 advancing the theory and application of PRMs across computer vision, healthcare (multi-class ECG), materials science, precision agriculture, ocean/climate monitoring, and structural engineering; several under review, one published at an IEEE-indexed conference, plus a pending high-entropy-alloy design patent.
- Delivered production-oriented results: interpretable multi-variable ocean models running in 1.09 ms on an ESP32, a 657 KB Sentinel-1 oil-spill detector at 11,500 tiles/s, and a 123M-parameter ECG classifier compressed 99.7% (1.3 MB) with negligible loss.

Course Team Leader

Jan 2025 – Present

Higher Colleges of Technology, Abu Dhabi, UAE

- Managing and guiding a team of lecturers across multiple HCT campuses for the System Architecture & Integration course.
- Developing and maintaining course materials, assessments, and standardised delivery frameworks.

Assistant Professor, Computer Information Science

Jan 2024 – Present

Higher Colleges of Technology, Abu Dhabi, UAE

- Courses: Fundamentals of Programming (Java), System Architecture & Integration, IoT & Security, Data Structures & Algorithms, Statistics & Probability.

Director of Software Engineering

Sep – Dec 2023

Plat.AI, Yerevan

Faculty Lecturer (Contract)

Jan 2023 – Dec 2023

Université française en Arménie (UFAR), Yerevan

Visiting Lecturer (Contract)

Aug 2022 – Jan 2023

American University of Armenia, Yerevan

Chief Executive Officer

Jun 2022 – Jul 2023

System View Armenia, Yerevan

- Developed LLVM/Clang-based auto-vectoriser for Xilinx ACAP (Visual System Integrator toolchain); peak speedup 2009× on GPU, 32.1× on CPU.

Associate Software Development Manager

Oct 2021 – Jun 2022

HelpSystems Armenia, Yerevan

Senior Lecturer (Contract)

Feb 2021 – Dec 2023

Yerevan State University, Faculty of Informatics & Applied Mathematics

Founder & Lecturer

Jan 2019 – Jun 2023

MutEx School, Yerevan

Software Architect (Contract)

2018 – 2020

Priotix, Yerevan · smart-home IoT system

Principal Software Engineer / Team Lead <i>HelpSystems Armenia, Yerevan</i>	Feb 2017 – Oct 2021
Principal / Senior Software Developer <i>Optym, Yerevan · logistics optimisation</i>	Aug 2015 – Feb 2017
Senior Software Developer <i>ArdInnoTekh LLC, Yerevan</i>	Aug 2013 – Jul 2015
Lecturer <i>Armsoft CS Learning Center, Yerevan</i>	Sep 2013 – Dec 2019
Head of System Programming Laboratory <i>Russian-Armenian University / ISP RAS, Yerevan</i>	Jan 2009 – Jul 2013
Led RAS System Programming Lab; directed Linux Foundation verification of system libraries and kernel modules; managed a team of researchers and engineers.	
Senior Lecturer <i>Russian-Armenian (Slavonic) University, Yerevan</i>	Feb 2007 – 2014
Software Developer <i>ISMOTech, Yerevan · firmware, C++/C#, ASP.NET</i>	Oct 2006 – Dec 2008
Software Developer <i>Lycos Europe · PHP/MySQL</i>	Mar 2005 – Oct 2006
Junior Research Scientist <i>Institute of Mathematics, National Academy of Sciences of RA</i>	2003 – 2005
Parallelisation of numerical algorithms for integral equations using MPI.	

Selected Applied Projects

Satellite Oil-Spill Detection <i>ExplainableAI.app</i>	2026
Built an interpretable PCA + sparse-polynomial detector for Sentinel-1 Deep-SAR imagery that outperformed a raw-image CNN baseline (G-mean $0.606 \rightarrow 0.822$); a 657 KB model running at $\approx 87 \mu\text{s}/\text{tile}$ ($\sim 11,500$ tiles/s) enabling on-board triage with a 76 KB downlink for 3,163 flagged tiles.	
Food-Security & Agricultural Forecasting Corridor <i>ExplainableAI.app</i>	2026
Deployed interpretable models for soil-crop suitability (accuracy 0.948), crop-yield forecasting ($R^2 = 0.656$), and agricultural commodity prices ($R^2 = 0.839$, up to 0.929 with a price anchor) as low-footprint, auditable alternatives to black-box pipelines.	
On-Device Ocean Profiling (TinyML) <i>ExplainableAI.app</i>	2026
Designed quantized (int16) polynomial models for multi-variable ocean profiling that run in 1.09 ms on an ESP32 and 1.49 ms on an RP2350 at $180 \mu\text{J}/\text{inference}$, targeting autonomous marine platforms with no cloud dependency.	

Publications

Applied Machine Learning & Explainable AI

- **Martirosyan, V.** “Accuracy and Interpretability from a Single Solver: Polynomial Regression up to Degree 5 for Concrete Compressive Strength Prediction.” *ICACTE 2026 (published)*. ($R^2 = 0.859$, RMSE 6.25 MPa; 74% of retained terms match known concrete physics.)
- **Martirosyan, V.** “Efficient Multi-Class ECG Classification Using Polynomial Regression with an Analytical Solver.” *Applied Computational Intelligence and Soft Computing (major revision)*, 2026. ($82.75\% \pm 1.03\%$ six-class accuracy; 12 μ s inference; beats CNN, RF, XGBoost and SVM baselines.)
- **Martirosyan, V.** “Automated Multi-Class ECG Arrhythmia Classification Using an Ensemble of Polynomial Regression Models.” *IEEE ICCA 2026 (submitted)*, 2026. (78.32% balanced accuracy; 123.1M parameters compressed 99.7% to 1.3 MB within ≈ 1 pp.)
- **Martirosyan, V.** “Sparse Polynomial Regression for Interpretable Yield-Strength Prediction in High-Entropy Alloys.” *IEEE ITT 2026 (under review)*. ($R^2 = 0.738$, tied with RF/XGBoost and above a neural network; 8/11 terms map to known hardening mechanisms.)
- **Martirosyan, V.** “Efficient Crop Type Mapping via Degree-2 Polynomial Feature Expansion of Multi-Temporal Optical and SAR Data.” *IEEE TGRS (submitted)*. (98.67% accuracy; 2.3 s training on 16 cores; $17\times$ SIMD speedup.)
- **Martirosyan, V.** “Stagewise Regularized Polynomial Regression: A Hybrid Optimization Strategy for Medical Diagnosis on Small Datasets.” Feb 2026. (86.7% accuracy on UCI Cleveland Heart Disease; 93% model compression; ≈ 36 FLOPs/prediction.)
- **Martirosyan, V.** “Heart Disease Diagnosis Using Third-Order Polynomial Regression Models with L2 Regularization.” Feb 2026. (81.8% accuracy, 10-fold CV; full interpretability via explicit coefficients.)
- **Martirosyan, V.** “Polynomial Regression Models for Interpretable Machine Learning: Theory, Optimization, and Applications.” Feb 2026.
- **Martirosyan, V.** “Explainable and Efficient Digit Classification via Polynomial Regression Networks: A Transparent Alternative to Deep Learning.” arXiv preprint, Jan 2026. (96.85% average MNIST accuracy; 84% parameter reduction at $<1\%$ loss.)

Edge AI, Ocean & Climate Monitoring

- **Martirosyan, V.** “On-Device TinyML for Multi-Variable Ocean Profiling” (eICocean). *eICocean 2027 (under review)*. ($R^2 = 0.973$ temperature / 0.794 salinity / 0.895 dissolved oxygen; 4.4 μ s desktop, 1.09 ms ESP32, 1.49 ms RP2350; 180 μ J/inference.)
- **Martirosyan, V.** “Interpretable Polynomial Zone Ensembles for Global Ocean Temperature and Salinity Profile Reconstruction.” *Q1 journal (under review)* and *IEEE JSTARS (prepared)*. (534,090 WOA 2018 samples; $R^2 = 0.9703 / 0.8376$; sub-ms, sub-MB.)
- **Martirosyan, V.** “Depth-Zone Polynomial Regression for Global Ocean Climatology.” *IEEE journal (under review)*.
- **Martirosyan, V.** “Auditable Kilobyte-Scale Surrogates for the Global Significant-Wave-Height Climatology.” *IEEE JSTARS (prepared)*. ($R^2 = 0.945$, RMSE 0.262 m; coastal R^2 0.34 $\rightarrow$ 0.52 with in-situ buoys.)
- **Martirosyan, V.** “Blended Ocean Visualization: Combining Sparse Polynomial Regression with Residual Fields.” Working paper, 2026.

HPC, Compilers & Systems

- **Martirosyan, V.** “LEAP: Lifting Legacy x86-64 Binaries to Recompilable IR for Automatic Parallelization.” *Frontiers of Computer Science (submitted)*, 2026.
(100%/98% liftability across 43 programs; $10.3\times$ geomean speedup over -00; $246\times$ peak GPU of-fload; 377/377 tests pass.)
- **Martirosyan, V.** “HPC++: An LLVM-Based Automatic Parallelization Framework with Heterogeneous CPU-GPU Execution.” Feb 2026.
(Peak speedup $2009\times$ on GPU; $32.1\times$ on CPU; 7 strategies, zero source modification.)
- **Martirosyan, V.** et al. “Testing of Linux File System Drivers.” *Proceedings of ISP RAS*, vol. 23, Moscow, Dec 2012.
- **Martirosyan, V.** et al. “The Spruce System: Quality Verification of Linux File System Drivers.” *SYRCoSE 2012*, May 2012.
- **Martirosyan, V.** et al. “Dynamic Verification of Linux File System Drivers.” *CSIT 2011*, Sep 2011.
- **Martirosyan, V.** et al. “The Spruce System: Linux Kernel Module Verification Automation” (in Russian). *7th Annual Conference of RAU*, Dec 2012.
- **Martirosyan, V.** et al. “Fault Simulation in Linux Kernel Module Verification” (in Russian). *7th Annual Conference of RAU*, Dec 2012.
- **Martirosyan, V.** et al. “Linux Verification – Challenge of Today.” *7th Annual Conference of RAU*, Dec 2012.
- **Martirosyan, V.** et al. “X2C – XML To Code: Advantages of Standards.” *4th Scientific Annual Conference of RAU*, Dec 2009.
- **Martirosyan, V.** et al. “X2C: Code Generation Technology.”

Books & Tutorials

- **Martirosyan, V.** *Programming via Operating Systems, Second Edition*. Hilfmann Press, Sep 2024.
- **Martirosyan, V.** et al. *Tutorial: Basics of Programming in OS Linux Environment* (in Russian). MIPT, Apr 2011.
- **Martirosyan, V.** et al. *Tutorial: Programming in GNU/Linux Environment – From System Calls to the GUI* (in Russian). RAU Publishing, Dec 2010.

Translations

- *Jinnetic Engineering* by Richard Stallman, translated into Armenian. Dec 2010.

Education

Ph.D., Computational Mathematics

2006 – 2009

Yerevan State University

Dissertation: “Construction of an Algorithm for the Pseudoinversion of Bidiagonal Matrices and an Investigation of its Parallelization”

M.Sc., Informatics and Applied Mathematics

2004 – 2006

Yerevan State University

Professional Development & Certifications

AI, Machine Learning & Data Science

- **Fundamentals of Deep Learning** *NVIDIA*, Apr 2025
- Advanced NoSQL for Data Science *LinkedIn*, Jan 2025
- Statistics Foundations 1: The Basics *LinkedIn*, Sep 2024
- Artificial Intelligence Foundations: Thinking Machines *LinkedIn*, Jul 2024
- Introduction to Artificial Intelligence *LinkedIn*, Jul 2024

Leadership & Communication

- How to Be an Effective Manager to Drive Impact *LinkedIn*, Jul 2024
- Storytelling to Influence Leadership and Decision Makers *LinkedIn*, May 2024
- Improving Your Thinking *LinkedIn*, Jul 2024
- Leading with Emotional Intelligence *LinkedIn*, Apr 2019
- Public Speaking Foundations *LinkedIn*, Feb 2019

Embedded Systems, FPGA & Hardware

- C Programming for Embedded Applications *LinkedIn*, Nov 2024
- Introduction to FreeRTOS and Basic Task Management *LinkedIn*, Nov 2024
- Getting Started with RISC-V *LinkedIn*, Jul 2024
- Learning Verilog for FPGA Development *LinkedIn*, Jul 2024
- Learning FPGA Development *LinkedIn*, Jul 2024
- Introduction to Quantum Computing *LinkedIn*, Jul 2024

Software Architecture & Engineering

- Software Architecture: Domain-Driven Design *LinkedIn*, Jul 2024
- Microservices Foundations *LinkedIn*, Jul 2024
- Microservices: Asynchronous Messaging *LinkedIn*, Jul 2024
- Microservices: Security *LinkedIn*, Jul 2024